Soon after lift-off from the Kennedy Space Center in Florida, USA, Shukla also experienced the tremendous gravitational force or G-force that Rakesh Sharma, India's first astronaut, talked about when his Soviet Soyuz T-11 spacecraft took him to space in April 1984. Sharma recalled, "Because you are seated in the spacecraft looking upwards for lift-off, the G-force that you feel on your rib cage is four times the amount you feel on your waist. It presses against the spine, leaving very little space for the lungs to expand. So, breathing becomes difficult."

By launch day on June 25 , Shukla was only too eager to get off the ground. He had waited a month in quarantine and then faced several postponements of the Crew Dragon's launch for technical reasons. "When I was sitting in the capsule on the launch pad," he says, "the only thought on my mind was: let's just go this time. When the ride started, I kept getting pushed back strongly in the seat. Then, suddenly, I felt nothing. There was silence and we were floating in vacuum. What a ride—it was amazing." Far below, in the computer-lined control room of Launch Complex 39A, his mother, Asha Shukla, who anxiously watched the spacecraft's ascent on a screen, wiped the tears from her eyes and broke out into a huge smile of joy and relief.

When he reached the ISS, Shukla brushed aside the motion sickness and smiled a lot. Along with his three team members on Axiom Mission 4, he hugged the seven astronauts who had come to the space station earlier. Shukla joked, "I am learning like a baby, how to walk, speak and eat in space." Ax-4 mission commander Peggy Whitson fixed a silver pin on the lapel of his deep blue overalls, designating Shukla as astronaut number 634—the 634th human to have orbited in space. The ISS is a multinational collaborative effort, involving the US, Russia, Europe, Japan and Canada, and was built in phases from 1998 to 2011. It is now an orbiting space laboratory the size of four tennis courts. So far, 280 astronauts from 23 countries had spent time at the ISS. Shukla is the first Indian astronaut to live on the ISS and only the second Indian to go into orbit after Rakesh Sharma's historic feat 41 years ago. Aware of the weight of the moment, Shukla said, "It is a privilege to be among the few who have had the chance to see Earth from this vantage point."

Two days later, when Prime Minister Narendra Modi spoke to him, Shukla had to strap his legs to the ground to stop bobbing around. How was the view from ISS, Modi asked. Shukla's words were as resonant as Sharma's in 1984. "The first thought was the feeling of oneness of the Earth—there were no boundary lines or borders of countries," he said. "The second was when I saw India for the first time. It looked very big and grand, not like the 2-D paper maps of it." A smiling Modi then told Shukla that he is giving him some homework, saying, "We have to take Mission Gaganyaan (India's indigenous human space exploration programme) forward, we must build our own space station and also have to land Indian astronauts on the moon. Your experiences will be very helpful for these missions."



MEN ON A MISSION
From left, Wearing dark suits, members of the Ax-4 Mission relax after reaching the ISS: Slawosz Uznanski-Wisniewski (Poland), Peggy Whitson (US), Shukla and Tibor Kapu (Hungary), Jun. 26

Shukla's trip to ISS cost Rs 500 crore, but it will give India the exposure and experience needed for human interface processes in its own crewed spaceflight missions

NO FREE RIDE

Modi was also disarmingly telling the Indian Space Research Organisation (ISRO), which sponsored Shukla's Rs 500 crore trip, that there were no free lunches or, in this case, space rides. Not that ISRO needed much goading. In